

SIM1003 Calculus II
Tutorial 7

All even-numbered questions in the following exercises

Exercise 10.1:

Finding Cartesian from Parametric Equations

Exercises 1–18 give parametric equations and parameter intervals for the motion of a particle in the xy -plane. Identify the particle's path by finding a Cartesian equation for it. Graph the Cartesian equation and indicate the portion of the graph traced by the particle and the direction of motion.

1. $x = 3t, \quad y = 9t^2, \quad -\infty < t < \infty$
2. $x = -\sqrt{t}, \quad y = t, \quad t \geq 0$
3. $x = 2t - 5, \quad y = 4t - 7, \quad -\infty < t < \infty$
4. $x = 3 - 3t, \quad y = 2t, \quad 0 \leq t \leq 1$
5. $x = \cos 2t, \quad y = \sin 2t, \quad 0 \leq t \leq \pi$
6. $x = \cos(\pi - t), \quad y = \sin(\pi - t), \quad 0 \leq t \leq \pi$
7. $x = 4 \cos t, \quad y = 2 \sin t, \quad 0 \leq t \leq 2\pi$
8. $x = 4 \sin t, \quad y = 5 \cos t, \quad 0 \leq t \leq 2\pi$
9. $x = \sin t, \quad y = \cos 2t, \quad -\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$
10. $x = 1 + \sin t, \quad y = \cos t - 2, \quad 0 \leq t \leq \pi$
11. $x = t^2, \quad y = t^6 - 2t^4, \quad -\infty < t < \infty$
12. $x = \frac{t}{t-1}, \quad y = \frac{t-2}{t+1}, \quad -1 < t < 1$
13. $x = t, \quad y = \sqrt{1-t^2}, \quad -1 \leq t \leq 0$
14. $x = \sqrt{t+1}, \quad y = \sqrt{t}, \quad t \geq 0$
15. $x = \sec^2 t - 1, \quad y = \tan t, \quad -\pi/2 < t < \pi/2$
16. $x = -\sec t, \quad y = \tan t, \quad -\pi/2 < t < \pi/2$
17. $x = -\cosh t, \quad y = \sinh t, \quad -\infty < t < \infty$
18. $x = 2 \sinh t, \quad y = 2 \cosh t, \quad -\infty < t < \infty$

Exercise 10.2:

Tangent Lines to Parametrized Curves

In Exercises 1–14, find an equation for the line tangent to the curve at the point defined by the given value of t . Also, find the value of d^2y/dx^2 at this point.

1. $x = 2 \cos t, y = 2 \sin t, t = \pi/4$

2. $x = \sin 2\pi t, y = \cos 2\pi t, t = -1/6$

3. $x = 4 \sin t, y = 2 \cos t, t = \pi/4$

4. $x = \cos t, y = \sqrt{3} \cos t, t = 2\pi/3$

5. $x = t, y = \sqrt{t}, t = 1/4$

6. $x = \sec^2 t - 1, y = \tan t, t = -\pi/4$

7. $x = \sec t, y = \tan t, t = \pi/6$

8. $x = -\sqrt{t+1}, y = \sqrt{3t}, t = 3$

9. $x = 2t^2 + 3, y = t^4, t = -1$

10. $x = 1/t, y = -2 + \ln t, t = 1$

11. $x = t - \sin t, y = 1 - \cos t, t = \pi/3$

12. $x = \cos t, y = 1 + \sin t, t = \pi/2$

13. $x = \frac{1}{t+1}, y = \frac{t}{t-1}, t = 2$

14. $x = t + e^t, y = 1 - e^t, t = 0$

Implicitly Defined Parametrizations

Assuming that the equations in Exercises 15–20 define x and y implicitly as differentiable functions $x = f(t)$, $y = g(t)$, find the slope of the curve $x = f(t)$, $y = g(t)$ at the given value of t .

15. $x^3 + 2t^2 = 9, 2y^3 - 3t^2 = 4, t = 2$

16. $x = \sqrt{5 - \sqrt{t}}, y(t - 1) = \sqrt{t}, t = 4$

17. $x + 2x^{3/2} = t^2 + t, y\sqrt{t + 1} + 2t\sqrt{y} = 4, t = 0$

18. $x \sin t + 2x = t, t \sin t - 2t = y, t = \pi$

19. $x = t^3 + t, y + 2t^3 = 2x + t^2, t = 1$

20. $t = \ln(x - t), y = te^t, t = 0$

Area

21. Find the area under one arch of the cycloid $x = a(t - \sin t)$, $y = a(1 - \cos t)$.

22. Find the area enclosed by the y -axis and the curve $x = t - t^2$, $y = 1 + e^{-t}$.

23. Find the area enclosed by the ellipse $x = a \cos t$, $y = b \sin t$, $0 \leq t \leq 2\pi$.

24. Find the area under $y = x^3$ over $[0, 1]$ using the following parametrizations.

a. $x = t^2$, $y = t^6$

b. $x = t^3$, $y = t^9$

Lengths of Curves

Find the lengths of the curves in Exercises 25–30.

25. $x = \cos t$, $y = t + \sin t$, $0 \leq t \leq \pi$

26. $x = t^3$, $y = 3t^2/2$, $0 \leq t \leq \sqrt{3}$

27. $x = t^2/2$, $y = 2(t+1)^{3/2}/3$, $0 \leq t \leq 4$

28. $x = (2t+3)^{3/2}/3$, $y = t + t^2/2$, $0 \leq t \leq 3$

29. $x = 8 \cos t + 8t \sin t$, $y = 8 \sin t - 8t \cos t$, $0 \leq t \leq \pi/2$

30. $x = \ln(\sec t + \tan t) - \sin t$, $y = \cos t$, $0 \leq t \leq \pi/3$

Exercise 10.3:

Polar Coordinates

2. Which polar coordinate pairs label the same point?
- | | | |
|-------------------------|-------------------|-------------------|
| a. $(-2, \pi/3)$ | b. $(2, -\pi/3)$ | c. (r, θ) |
| d. $(r, \theta + \pi)$ | e. $(-r, \theta)$ | f. $(2, -2\pi/3)$ |
| g. $(-r, \theta + \pi)$ | h. $(-2, 2\pi/3)$ | |
3. Plot the following points (given in polar coordinates). Then find all the polar coordinates of each point.
- | | |
|------------------|--------------|
| a. $(2, \pi/2)$ | b. $(2, 0)$ |
| c. $(-2, \pi/2)$ | d. $(-2, 0)$ |
4. Plot the following points (given in polar coordinates). Then find all the polar coordinates of each point.
- | | |
|------------------|-------------------|
| a. $(3, \pi/4)$ | b. $(-3, \pi/4)$ |
| c. $(3, -\pi/4)$ | d. $(-3, -\pi/4)$ |

Graphing Sets of Polar Coordinate Points

Graph the sets of points whose polar coordinates satisfy the equations and inequalities in Exercises 11–26.

- | | |
|---|--|
| 11. $r = 2$ | 12. $0 \leq r \leq 2$ |
| 13. $r \geq 1$ | 14. $1 \leq r \leq 2$ |
| 15. $0 \leq \theta \leq \pi/6, \quad r \geq 0$ | 16. $\theta = 2\pi/3, \quad r \leq -2$ |
| 17. $\theta = \pi/3, \quad -1 \leq r \leq 3$ | 18. $\theta = 11\pi/4, \quad r \geq -1$ |
| 19. $\theta = \pi/2, \quad r \geq 0$ | 20. $\theta = \pi/2, \quad r \leq 0$ |
| 21. $0 \leq \theta \leq \pi, \quad r = 1$ | 22. $0 \leq \theta \leq \pi, \quad r = -1$ |
| 23. $\pi/4 \leq \theta \leq 3\pi/4, \quad 0 \leq r \leq 1$ | |
| 24. $-\pi/4 \leq \theta \leq \pi/4, \quad -1 \leq r \leq 1$ | |
| 25. $-\pi/2 \leq \theta \leq \pi/2, \quad 1 \leq r \leq 2$ | |
| 26. $0 \leq \theta \leq \pi/2, \quad 1 \leq r \leq 2$ | |

Polar to Cartesian Equations

Replace the polar equations in Exercises 27–52 with equivalent Cartesian equations. Then describe or identify the graph.

27. $r \cos \theta = 2$

28. $r \sin \theta = -1$

29. $r \sin \theta = 0$

30. $r \cos \theta = 0$

31. $r = 4 \csc \theta$

32. $r = -3 \sec \theta$

33. $r \cos \theta + r \sin \theta = 1$

34. $r \sin \theta = r \cos \theta$

35. $r^2 = 1$

36. $r^2 = 4r \sin \theta$

37. $r = \frac{5}{\sin \theta - 2 \cos \theta}$

38. $r^2 \sin 2\theta = 2$

39. $r = \cot \theta \csc \theta$

40. $r = 4 \tan \theta \sec \theta$

41. $r = \csc \theta e^{r \cos \theta}$

42. $r \sin \theta = \ln r + \ln \cos \theta$

43. $r^2 + 2r^2 \cos \theta \sin \theta = 1$

44. $\cos^2 \theta = \sin^2 \theta$

45. $r^2 = -4r \cos \theta$

46. $r^2 = -6r \sin \theta$

47. $r = 8 \sin \theta$

48. $r = 3 \cos \theta$

49. $r = 2 \cos \theta + 2 \sin \theta$

50. $r = 2 \cos \theta - \sin \theta$

51. $r \sin \left(\theta + \frac{\pi}{6} \right) = 2$

52. $r \sin \left(\frac{2\pi}{3} - \theta \right) = 5$

Cartesian to Polar Equations

Replace the Cartesian equations in Exercises 53–66 with equivalent polar equations.

53. $x = 7$

54. $y = 1$

55. $x = y$

56. $x - y = 3$

57. $x^2 + y^2 = 4$

58. $x^2 - y^2 = 1$

59. $\frac{x^2}{9} + \frac{y^2}{4} = 1$

60. $xy = 2$

61. $y^2 = 4x$

62. $x^2 + xy + y^2 = 1$

63. $x^2 + (y - 2)^2 = 4$

64. $(x - 5)^2 + y^2 = 25$

65. $(x - 3)^2 + (y + 1)^2 = 4$

66. $(x + 2)^2 + (y - 5)^2 = 16$